Semantic Cooperative Network for Few-Shot Remote Sensing Object Detection

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***Abstract*—Few-shot remote sensing object detection aims to recognize novel-class objects using limited annotated samples. However, the scarcity of novel-class samples makes the model highly dependent on base-class knowledge, while category ambiguity in remote sensing scenes further amplifies this bias. Semantic cooperative network (SCNet) is a novel few-shot remote sensing object detection framework. This framework establishes a cooperative mechanism between visual class-relation modeling and dynamic semantic priors to mine and enhance discriminative visual cues related to novel classes. Specifically, the class-relation awakening module (CAM) models contextual dependencies among class prototypes and adaptively refines RoI features using the activated relational information, thereby enhancing interclass separability. In addition, the dual-prompt semantic module (DSM) constructs adaptive semantic priors aligned with visual features using complementary prompts, which helps reduce the dependence of novel-class representations on base-class knowledge. Extensive experiments on the DIOR, NWPU VHR-10, and RSOD benchmark datasets demonstrate that the proposed method achieves performance comparable to various state-of-the-art methods under different shot settings.**

***Index Terms*—Class-aware relational modeling, cross-modal learning, dynamic semantic guidance, few-shot object detection (FSOD).**

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I. Introduction

REMOTE sensing object detection has become a research topic of broad interest in recent years, driven by continuous advances in satellite technology and the growing availability of large-scale remote sensing data [1], [2], [3], [4], [5], [6]. With the rapid development of deep learning techniques [7], [8], [9], [10], [11], object detection algorithms have achieved remarkable progress. For instance, excellent detection results have been reported by representative detectors, including Faster R-CNN [12] and YOLO [13], on multiple remote sensing image datasets (RSIs). Nevertheless, such approaches generally require substantial volumes of precisely labeled training data. For remote sensing images, the annotation process not only requires domain expertise but also involves substantial human effort, time, and economic cost. Therefore, few-shot object detection (FSOD) in remote sensing scenarios has gradually attracted extensive attention from researchers and has become a research hotspot [14], [15], [16], [17].

However, applying FSOD to remote sensing scenarios still faces two key challenges. First, limited novel-class samples cannot adequately cover the common orientation and scale variations of remote sensing objects. Consequently, during fine-tuning, the model tends to rely on the visual patterns learned from base classes, leading to base-class-biased representations of novel classes [18]. Second, remote sensing objects commonly exhibit high interclass similarity and large intraclass variation, while their category discrimination is also susceptible to scene context [19], [20]. Different categories may share similar geometric structures, texture characteristics, or spatial environments, causing the model to confuse categories with similar visual appearances or scene semantics. As shown in Fig. 1, bridges and overpasses, as well as railway stations and highway toll stations, may exhibit highly similar visual structures, whereas objects from the same category, such as dams or stadiums, may vary considerably across imaging conditions and surrounding environments. These factors make it difficult for the model to learn stable and discriminative novel-class representations and establish clear category decision boundaries.

To address these issues, existing FSOD methods generally follow either meta-learning [21], [22], [23], [24] or transfer-learning paradigms [25], [26], [27], [28]. Meta-learning-based methods usually construct support–query interactions or prototype-guided representations to improve adaptation to novel classes. For example, P-CNN [29] enhances class

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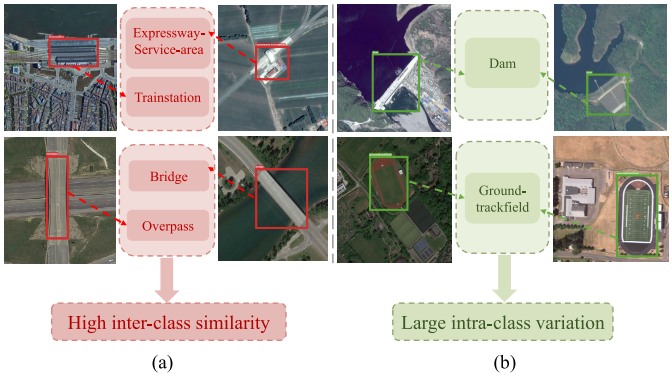


Fig. 1. Examples in remote sensing scenes. (a) Cases with high interclass similarity. In such scenarios, different categories often exhibit highly similar texture and structural characteristics due to imaging perspectives, spatial resolution limitations, and background interference. (b) Cases with large intraclass variation. Objects within the same category often show significant differences in shape, scale, and color, resulting in strong intraclass variability.

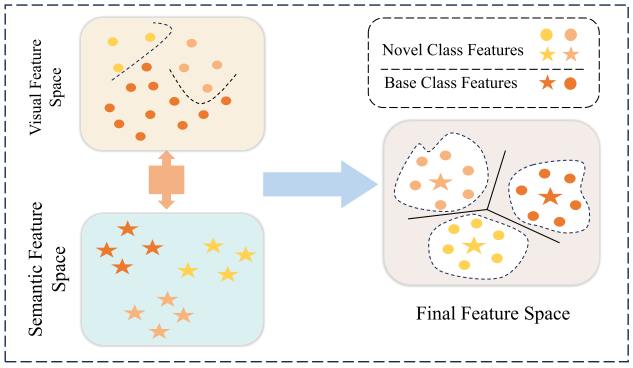


Fig. 2. Proposed framework employs an attention-constrained relation modeling strategy to enhance visual representations, thereby encouraging a more distinct feature-space separation between novel and base classes and mitigating the adverse effects arising from large intraclass variation and high interclass similarity. It further incorporates interaction with rich semantic features to compensate for insufficient feature representation of novel classes. Ultimately, the proposed method improves the representation capability of novel classes while enhancing their discriminability from other classes in the feature space.

discriminability through prototype learning, while MLII-FSOD [30] alleviates feature misalignment by exploiting multilevel feature interaction. However, these methods often require complex task construction and may still struggle when interclass differences are subtle. In contrast, transfer-learning-based methods first train detectors using abundant base-class samples and then fine-tune them with a few novel-class samples. After TFA [31] demonstrated the effectiveness of simple fine-tuning, this paradigm gradually became a practical solution for FSOD. Representative methods such as G-FSDet [32] and SAE-FSDet [33] improve few-shot detection by mitigating novel–base conflicts, maintaining label consistency, or improving proposal quality.

Despite these advances, existing methods still exhibit clear limitations in few-shot remote sensing scenarios. In terms of visual representation, metric learning, contrastive learning, and prototype-matching methods generally alleviate category confusion by improving intraclass compactness and interclass separability [34], [35]. However, most of them focus on optimizing individual RoI features or independently learned class representations, while making limited use of the relative relationships between target classes and their confusing classes. When class prototypes are estimated from only a few samples, they are prone to deviation. Relying solely on independent matching between an RoI and each class prototype therefore makes it difficult to establish clear and stable decision boundaries against the most interfering similar categories. In terms of semantic guidance, existing vision–language FSOD methods commonly use class names, manually constructed class descriptions, or a single form of learnable prompt to supplement limited visual information [36], [37], [38]. However, fixed class names or description prompts provide relatively stable class knowledge but cannot adapt effectively to the current few-shot task, whereas a single form of learnable prompt offers task adaptability but may lack stable, explicit, and comprehensive class-attribute knowledge. Therefore, existing methods remain unable to sufficiently exploit interclass relationships while simultaneously ensuring the task adaptability, semantic stability, and class discriminability of semantic priors.

To address these limitations, we propose the semantic cooperative network (SCNet) for few-shot remote sensing object detection, which jointly enhances novel-class representations through class-relation-aware visual feature refinement and dynamic semantic guidance. SCNet consists of two sequentially collaborative components, namely the class-relation awakening module (CAM) and the dual-prompt semantic module (DSM). CAM first models the contextual relationships among class prototypes and introduces class-relation information into each RoI feature through RoI–prototype interaction, thereby transforming the conventional independent RoI processing paradigm into class-relation-aware adaptive feature refinement. Meanwhile, the proposed attention margin separation loss (AMS) constrains the attention-response margin between the ground-truth class prototype and the strongest hard-negative class prototype, preventing incorrect categories with similar visual appearances or scene contexts from dominating the attention distribution and thus enhancing the separability of confusing categories. After relation-driven visual feature refinement by CAM, DSM jointly employs task-adaptive learnable prompts and description prompts that provide stable class knowledge. It further constructs dynamic semantic priors through domain adaptation, discriminative semantic interaction, and visual–semantic alignment. As shown in Fig. 2, these semantic priors collaborate with class-relation-aware visual features to supplement the limited visual information of novel classes, reduce the dependence of novel-class representations on base-class visual patterns, and enhance the representation and discrimination capabilities of the model for novel-class objects.

The main contributions of this work are summarized as follows.

1) SCNet is proposed as a remote sensing-oriented FSOD framework designed to enhance novel-class representations through class-aware relational modeling and dynamic semantic guidance.

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2) A CAM with an AMS is designed to refine RoI features by exploiting contextual relationships among class prototypes and suppressing confusing categories.
3) A DSM is developed to construct adaptive semantic priors and align them with visual features, thereby reducing base-class bias in novel-class representation learning.

II. Related Work

A. Few-Shot Object Detection

As research on FSOD continues to progress, numerous enhanced methods have been developed to mitigate problems including weakened model generalization and unsatisfactory detection accuracy arising from scarce training samples [39], [40], [41]. Meta-learning methods establish training paradigms that mimic low-shot tasks, allowing models to quickly adapt to unseen categories. For example, Meta-RCNN [42] achieves dynamic alignment of support and query features through feature reweighting, while FCT [43] further introduces a fully cross-Transformer structure to enhance cross-layer feature interaction, thereby improving matching performance in scenarios with large visual variations. To address the problem of insufficient feature representation, some studies have focused on improving feature modeling. VFA [44] constructs a class-agnostic feature aggregation mechanism and introduces variational distribution modeling, effectively enhancing the robustness of class representations and alleviating feature ambiguity across base-class and novel-class representations.

Transfer learning methods achieve rapid adaptation to novel classes by leveraging base-class knowledge. For example, Retentive R-CNN [45] alleviates the problem of catastrophic forgetting by designing a bias-balancing mechanism, while FM-FSOD [46] utilizes vision foundation models and large language models to enable feature transfer and inference without fine-tuning. These methods have collectively advanced the development of FSOD and gradually addressed the core challenges brought by limited samples [47], [48], [49], [50]. In remote sensing scenarios, objects commonly exhibit significant scale variations, arbitrary orientations, complex backgrounds, and high interclass visual similarity. These challenging imaging characteristics may exacerbate feature bias and semantic ambiguity under limited-sample conditions, making it difficult for the model to learn stable and discriminative class representations. Meanwhile, existing methods often enhance visual features or introduce semantic information independently, without sufficiently modeling class relationships and visual–semantic collaboration. Consequently, they still struggle to effectively alleviate category confusion and semantic inconsistency.

B. FSOD in Remote Sensing Images

Directly applying FSOD methods designed for natural scenes to remote sensing images usually leads to a significant performance degradation. This is mainly due to the characteristics commonly found in remote sensing images, such as large intraclass variation, small interclass differences, complex backgrounds, and sparse object distribution, which make feature representation and category discrimination more difficult. To address the above issues, recent studies have explored FSOD methods in remote sensing scenarios from different perspectives [51], [52], [53], [54], [55]. Some methods improve detection performance by enhancing multiscale feature representation capability. For example, MSOCL [56] uses a multiscale contrastive learning mechanism to strengthen feature consistency between objects and proposal regions, thereby improving feature discrimination in complex backgrounds. Another category of methods introduces self-training strategies to alleviate annotation scarcity. For instance, Zhang et al. [57] adopted a teacher–student framework to generate pseudo-labels to improve the recall of novel-class objects. MFKDIOD [58] combines knowledge distillation with implicit object mining, mitigating catastrophic forgetting while reducing the impact of false supervision caused by unlabeled objects. However, these methods still struggle to learn stable and discriminative class representations, which limits their generalization ability under extremely few-shot conditions [59], [60].

In recent years, related tasks such as remote sensing scene classification, semantic segmentation, and infrared small target detection have also been extensively investigated. Jia et al. [61] proposed a few-shot remote sensing scene classification method based on parameter-free attention and selective region matching, which identifies discriminative regions according to semantic similarity and spatial consistency, thereby alleviating complex background interference without introducing additional attention parameters. Sun et al. [62] integrated expert knowledge priors with hierarchical relational matching and improved generalization to unseen categories through structured semantic knowledge and contextual relation reasoning. Li et al. [63] jointly modeled spatial-domain and frequency-domain information to enhance the structural and discriminative representations of small and weak targets. For generalized few-shot semantic segmentation in remote sensing images, Jia et al. [64] investigated prototype updating and classifier adaptation from two complementary perspectives. The former improves the adaptability of novel-class prototypes by exploiting background information [64], whereas the latter jointly utilizes base-class classifier knowledge and query-image contextual information to narrow the capability gap between base-class and novel-class classifiers and enhance adaptation to intraclass variations [65]. These studies provide diverse insights into novel-class representation learning under few-shot remote sensing conditions from the perspectives of discriminative region mining, structured semantic modeling, feature representation enhancement, and prototype or classifier adaptation.

With the development of multimodal learning, FSOD methods based on vision–language models have gradually attracted attention by introducing prior knowledge to compensate for insufficient visual features under few-shot conditions. Liu et al. [36] construct textual semantic prototypes and integrate visual distillation with prompt tuning to achieve vision–language feature alignment, thereby alleviating the issues of insufficient category representation and cross-domain discrepancies under few-shot settings. SCG-FSOD [66] combines semantic

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knowledge distillation with CLIP text prior-driven supervised contrastive learning, effectively mitigating base-class bias. ST-FSOD [37] utilizes textual labels of each category to generate image-independent text-guided prototypes, thereby complementing sparse visual features. However, these methods typically use fixed textual prompts (e.g., "a photograph of an airplane") as input to the vision–language model CLIP [67] to generate semantic features. Nevertheless, such prompts remain unchanged throughout the training process, making the generated textual features lack adaptability, which easily leads to overfitting to the training data and thus limits the generalization ability to novel-class objects.

Although existing methods have advanced few-shot remote sensing object detection from the perspectives of visual feature enhancement and semantic prior introduction, they still lack an effective semantic cooperative mechanism to facilitate deep interactions between visual and semantic information, making it difficult for semantic priors to sufficiently guide visual feature learning. To this end, we start from class-aware relational modeling and dynamic semantic learning, obtain richer contextual information through feature activation, and introduce dynamic semantic priors. Meanwhile, based on a visual–text semantic relation matrix, the alignment process between class-aware visual features and semantic priors is constrained and guided, enabling semantic knowledge to effectively guide visual feature learning, thereby enhancing the discriminability of novel-class representations.

III. Methodology

A. Problem Definition

In this article, following convention, the standard settings in previous remote sensing image FSOD studies [29], [31] are followed. All dataset categories are partitioned into base classes C_{base} and novel classes C_{novel}, with $C_{base} \cap C_{novel} = \emptyset$. Specifically, each category in C_{base} contains abundant annotated data, while each category in C_{novel} has only K annotated instances. In addition, all annotated samples from C_{base} form the base dataset D_{base}, which can be represented as $D_{base} = \{(x_i, y_i), y_i \in C_{base}\}$, where x_i denotes an image sample from the base set, and y_i indicates its associated label. The annotated samples from C_{novel} form the novel dataset D_{novel}. Based on the above data setting, we adopt a two-stage fine-tuning transfer learning method for training. In the base training stage, D_{base} is used to train a base model. Then, in the fine-tuning phase, K object samples are evenly selected for every category within D_{base} and D_{novel} to form a class-balanced training set for model fine-tuning, allowing the model to adapt to novel-category object detection.

B. Overall Framework

In this article, the proposed SCNet uses SAE-FSDet [33] as the baseline. The base training stage follows the same procedure as the baseline, while the fine-tuning stage innovatively introduces two key components, CAM and DSM, as shown in Fig. 3. The two modules are sequentially integrated into the detection pipeline following a class-relation-driven visual refinement and dual-prompt-guided semantic enhancement scheme. Specifically, class prototypes are first constructed from the RoI features, and their contextual relationships are modeled to generate class-relation-aware representations through RoI–prototype cross-attention. The AMS loss strengthens the response to the ground-truth class prototype while suppressing interference from confusing classes, whereas residual fusion preserves the localization and structural information of the original RoI features. The refined features are then directly used for bounding-box regression and further enhanced by dynamic semantic priors constructed from learnable prompts and class-description prompts before classification. Through visual–semantic alignment and fusion, the resulting representations are used for final category prediction. In this way, CAM and DSM form a sequential and complementary pipeline that alleviates category confusion and base-class bias. Therefore, the total training loss of SCNet during the fine-tuning stage is defined as follows:

$$L = L_{rpn} + L_{reg} + L_{cls} + \alpha L_{AMS} + \beta L_{VSCA} \tag{1}$$

where L_{rpn}, L_{reg}, and L_{cls} denote the RPN loss, regression loss, and classification loss of the baseline during the fine-tuning stage, respectively, and α and β are hyperparameters that balance the contributions of the AMS and the visual-semantic contrastive alignment loss (VSCA).

C. Class-Relation Awakening Module

In remote sensing FSOD, RoI features are easily biased toward visually similar categories due to limited novel-class supervision. Since class prototypes contain category-level representative information, the contextual relationships among them can provide effective cues for distinguishing ambiguous category boundaries [60]. Based on this motivation, CAM performs class-aware relational modeling among class prototypes and uses the learned prototype correlations to refine RoI features. Specifically, through the attention margin separation mechanism, CAM encourages RoI features to focus on target-related prototypes while suppressing the influence of confusing prototypes. Therefore, the refined RoI features obtain stronger category-discriminative capability, improving the separability of visually similar categories. Fig. 3(b) illustrates the detailed architecture of CAM.

We regard the network mapping of the base detector as a trainable feature learner f_θ parameterized by θ, which is connected to the final classification and regression layers. The RoI feature of each instance is computed by $f_\theta : \mathbb{R}^D \rightarrow \mathbb{R}^M$. During training, category prototypes are dynamically constructed by aggregating ground-truth-labeled RoI features within the current mini-batch

$$\boldsymbol{p}_k = \frac{1}{|X_k|} \sum_{(\boldsymbol{x}_i, y_i) \in X_k} f_\theta(\boldsymbol{x}_i) \tag{2}$$

$$\boldsymbol{P} = \left[\boldsymbol{p}_1, \boldsymbol{p}_2, \ldots, \boldsymbol{p}_C\right]^T \tag{3}$$

where $\boldsymbol{x}_i$ denotes the D-dimensional feature representation of an RoI sample, y_i is its ground-truth label, k denotes the class index, C is the total number of categories, and X_k denotes the RoI samples of class k in the current mini-batch. The prototypes are recomputed at each training iteration and used

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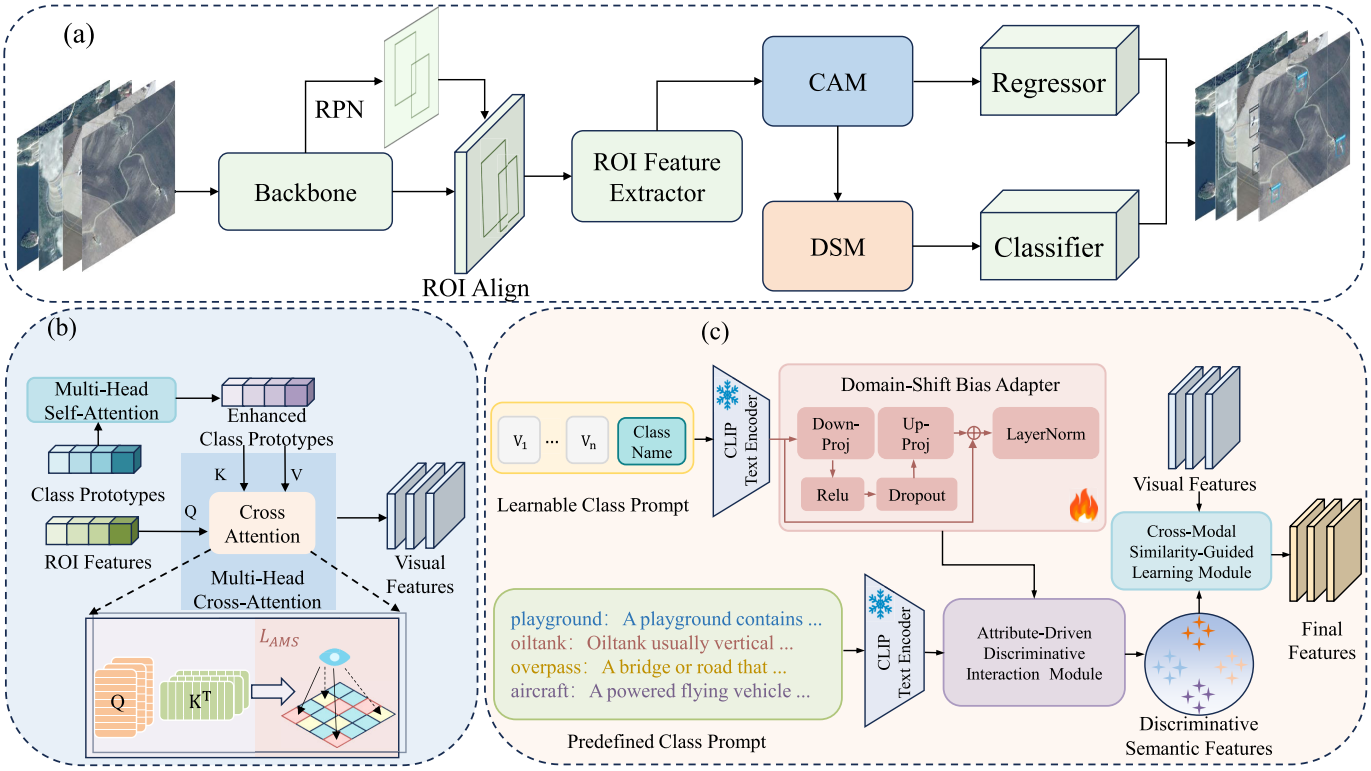


Fig. 3. Model architecture. (a) Overall architecture of SCNet, which is composed of two core modules, namely CAM and DSM. The RoI features are first transformed into class-aware visual features through CAM. The resulting features are subsequently passed to the regression branch for box regression and to DSM for semantic refinement, thereby enhancing the discriminative capability for novel classes and completing classification prediction. (b) Workflow of the CAM. The contextual correlations between RoI features and category prototypes are modeled through the attention-margin separation mechanism to generate class-aware visual features. (c) Workflow of the DSM. The semantic features obtained through the DBA, ADIM, and CSLM guide the learning and fusion of visual features.

only for class-relation modeling and RoI feature refinement, without moving-average updates or cross-iteration storage. During inference, category prototypes are neither constructed nor reused because ground-truth labels are unavailable; thus, no query-set or test-set annotations are involved.

To model the contextual relationships among different categories, a multihead self-attention mechanism is introduced to enhance the category prototypes, enabling each category prototype to be aware of the contextual information of other categories during the interaction process, thereby capturing potential information among categories. Through this strategy, the category prototypes are no longer independent representations, but form context-aware prototypes $\widetilde{\boldsymbol{P}}$ that contain category relationship information. The generation process is as follows:

$$\widetilde{\boldsymbol{P}} = \mathrm{Con}_{i=1}^{H}\left(\delta\left(\frac{\left(\boldsymbol{P}\boldsymbol{W}_i^{Q_p}\right)\left(\boldsymbol{P}\boldsymbol{W}_i^{K_p}\right)^T}{\sqrt{d_h}}\right)\boldsymbol{P}\boldsymbol{W}_i^{V_p}\right)\boldsymbol{W}^{O_p} \tag{4}$$

where H indicates the attention-head number, $\boldsymbol{W}_i^{Q_p}$, $\boldsymbol{W}_i^{K_p}$, and $\boldsymbol{W}_i^{V_p}$, respectively, represent the query, key, and value projection matrices of the ith head for the prototype features, $\boldsymbol{W}^{O_p}$ is the output projection matrix associated with the multihead attention module used to generate the enhanced category prototype, Con$(\cdot)$ denotes concatenation of outputs from all attention heads, and $\delta(\cdot)$ denotes the Softmax function.

On this basis, CAM further enables interaction modeling between RoI features and the enhanced category prototypes through a multihead cross-attention mechanism. In this process, related category information is adaptively aggregated through attention weights, thereby enabling RoI features to become more fully aware of the structural relationships and feature differences among different categories. In this way, the model can highlight discriminative features related to the target category while reducing the influence of irrelevant or distracting information on feature representation, allowing RoI features to obtain more fine-grained and expressive feature representations. The generation process of the final class-aware visual features $\widetilde{\boldsymbol{R}}$ is as follows:

$$\boldsymbol{R}_e = \mathrm{Con}_{i=1}^{H}\left(\delta\left(\frac{\left(\boldsymbol{R}\boldsymbol{W}_i^{Q_r}\right)\left(\widetilde{\boldsymbol{P}}\boldsymbol{W}_i^{K_r}\right)^T}{\sqrt{d_h}}\right)\left(\widetilde{\boldsymbol{P}}\boldsymbol{W}_i^{V_r}\right)\right)\boldsymbol{W}^{O_r} \tag{5}$$

$$\widetilde{\boldsymbol{R}} = (1-\lambda)\cdot\boldsymbol{R} + \lambda\cdot\boldsymbol{R}_e \tag{6}$$

where $\mathbf{W}_i^{Q_r}$ denotes the query projection matrix for RoI features, and $\mathbf{W}_i^{K_r}$ and $\mathbf{W}_i^{V_r}$ denote the key and value projection matrices for the enhanced category prototypes, respectively, $\boldsymbol{W}^{O_r}$ denotes the output projection matrix of the multihead attention for the class-aware visual features, $\boldsymbol{R}_e$ represents the class-aware enhanced features, $\boldsymbol{R}$ represents the RoI features, and λ is a hyperparameter that balances the fusion

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ratio between the original RoI features and the class-aware enhanced features.

In remote sensing scenarios, classification errors are usually dominated by a few interfering categories that are most similar to the target class, rather than the average influence of all negative categories. Therefore, the AMS loss enhances the attention response of RoI features to the correct class prototype while suppressing the response to the strongest confusing category, thereby improving the category-discriminative capability of RoI features. The specific process is as follows:

$$\boldsymbol{A} = \frac{1}{H}\sum_{i=1}^{H}\delta\left(\frac{\left(\boldsymbol{R}\boldsymbol{W}_i^{Q_r}\right)\left(\widetilde{\boldsymbol{P}}\boldsymbol{W}_i^{K_r}\right)^T}{\sqrt{d_h}}\right) \tag{7}$$

$$L_{AMS} = \frac{1}{N}\sum_{i=1}^{N}\max\left(0,\, m - \underbrace{\max_{j:y_j=y_i} A_{ij}}_{A_i^+} + \underbrace{\max_{j:y_j\neq y_i} A_{ij}}_{A_i^-}\right) \tag{8}$$

where $\boldsymbol{A}$ denotes the attention matrix generated by the multi-head cross-attention mechanism. For the ith RoI, A_i^+ denotes its maximum attention response to the ground-truth class prototype, while A_i^- denotes its maximum attention response to other class prototypes. By maximizing A_i^+ and restricting A_i^-, a predefined margin m is maintained between them, thereby forming a discriminative attention constraint.

D. Dual-Prompt Semantic Module

Since novel-class samples are limited, relying only on visual features easily makes novel-class representations affected by base-class knowledge. To this end, DSM constructs a dual-path semantic prior collaborative guidance mechanism and introduces dynamic semantic priors to guide novel-class representation learning, as shown in Fig. 3(c). Specifically, learnable class prompts generate flexible category-specific semantics through the domain-shift bias adapter (DBA), while class-description prompts provide relatively stable semantic priors. Subsequently, the attribute-driven discriminative interaction module (ADIM) fuses the two types of semantic prompts to generate more diverse and expressive semantic representations; the cross-modal similarity-guided learning module (CSLM) further aligns visual features with semantic priors, enabling semantic information to effectively guide visual feature learning, thereby weakening the reliance of representations for novel classes on knowledge learned from base classes and enhancing their discriminative capability for novel-class objects.

1) Category Prompts: To construct dynamic semantic priors suitable for novel remote sensing classes, DSM introduces both learnable class prompts and class-description prompts. The learnable class prompts consist of prompt vectors and class names and can be adaptively optimized according to few-shot tasks, thereby providing more task-adaptive semantic representations and alleviating the overfitting problem caused by fixed class prompts. The class-description prompts are constructed for remote sensing scenes, providing relatively stable and richer semantic constraints for semantic learning to compensate for the difficulty of fully expressing fine-grained semantics of remote sensing objects by relying only on class names. The two types of prompts form a complementary relationship, enabling the generated semantic priors to be both task-adaptive and capable of maintaining stable category semantic representations, thereby better serving novel-class representation learning in remote sensing. The semantic features $\boldsymbol{f}_k^{\text{learn}}$ and $\boldsymbol{f}_k^{\text{description}}$ of the kth class generated by the CLIP text encoder are obtained as follows:

$$P_k^{\text{learn}} = [\boldsymbol{V}_1]\,[\boldsymbol{V}_2]\ldots\left[\boldsymbol{V}_{n_p}\right]\,[\text{class}_k] \tag{9}$$

$$P_k^{\text{description}} = [w_k] \tag{10}$$

$$\boldsymbol{f}_k^{\text{learn}} = \text{Encoder}\left(P_k^{\text{learn}}\right) \tag{11}$$

$$\boldsymbol{f}_k^{\text{description}} = \text{Encoder}\left(P_k^{\text{description}}\right) \tag{12}$$

where $[\boldsymbol{V}_{n_p}]$ denotes the n_pth learnable prompt vector, n_p is the number of learnable prompt vectors, class_k denotes the class-label token associated with category k, w_k gives the text-modality description sentence for category k, and k indicates the class index.

2) Domain-Shift Bias Adapter: As the vision–language model CLIP is pretrained using natural image–text pairs, simply injecting learnable prompts into the text encoder of CLIP usually cannot adequately accommodate the domain-specific characteristics of remote sensing imagery, leading to insufficient semantic alignment between visual and textual modalities. Therefore, a DBA is designed to obtain aligned learnable text features $\widetilde{\boldsymbol{f}}^{\text{learn}}$. It is worth noting that text features generated from learnable prompts have stronger task adaptability, but are also more susceptible to domain shift in remote sensing, leading to semantic drift. In contrast, manually constructed class descriptions provide relatively stable semantic priors and serve as reliable semantic anchors for each category. Therefore, to avoid destroying the existing semantic structure in the description branch, only the learnable branch is adapted

$$\widetilde{\boldsymbol{f}} = \boldsymbol{f}^{\text{learn}} + \boldsymbol{W}_2\,\text{Dropout}\left(\text{ReLU}\left(\boldsymbol{W}_1\boldsymbol{f}^{\text{learn}}\right)\right) \tag{13}$$

$$\widetilde{\boldsymbol{f}}^{\text{learn}} = \text{LayerNorm}\left(\widetilde{\boldsymbol{f}}\right) \tag{14}$$

where $\boldsymbol{f}^{\text{learn}}$ denotes the learnable text features, $\boldsymbol{W}_1$ and $\boldsymbol{W}_2$ are the projection matrices in the feedforward network, LayerNorm$(\cdot)$ denotes layer normalization, and Dropout$(\cdot)$ indicates the dropout operation. The adapter with a residual-connected bottleneck effectively mitigates the domain bias in the feature vectors.

3) Attribute-Driven Discriminative Interaction Module: Learnable prompt features can dynamically adjust semantic expressions according to the object detection task, but their semantic stability is relatively limited; class-description text features contain more stable and complete category semantic priors, but lack task-specific adaptability. To fully exploit their complementary advantages, DSM designs an ADIM to construct a discriminative fusion mechanism for jointly modeling both categories of textual features, as illustrated in Fig. 4. ADIM introduces a discriminative semantic activation mechanism to highlight discriminative semantic dimensions related to the target category. Specifically, the class-description

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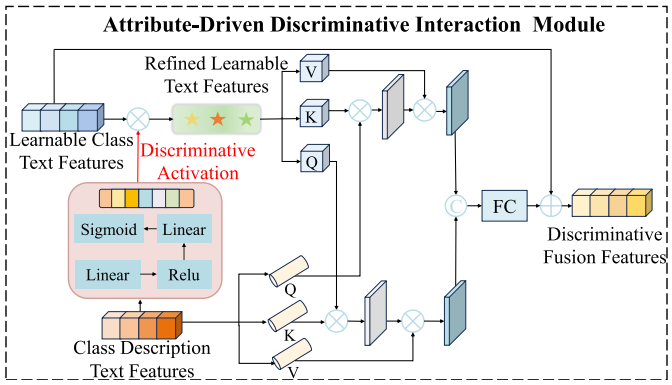


Fig. 4. Structure of the ADIM. The module generates discriminative semantic attention from class-description features to modulate learnable text features and further achieves complementary integration of the two types of semantic information through bidirectional interactive fusion.

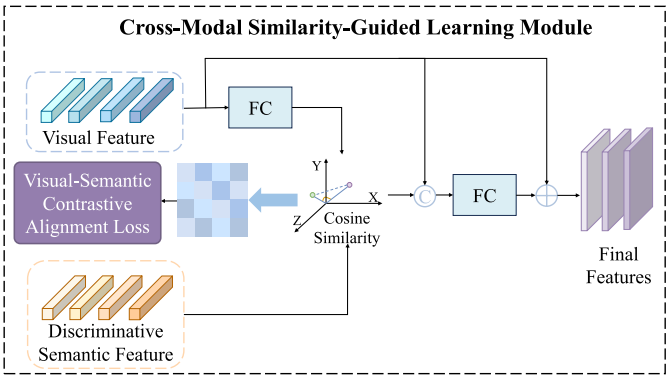


Fig. 5. Structure of the CSLM. This module performs semantic-guided refinement of region features by projecting visual representations onto a semantic embedding space and modeling visual–text relationships, and further strengthens the cross-modal semantic coherence of visual and textual representations through cross-modal contrastive learning.

semantic feature $\boldsymbol{f}^{\text{description}}$ generates discriminative semantic attention $\boldsymbol{A}_{\text{dis}}$ and adaptively modulates the learnable text feature $\widetilde{\boldsymbol{f}}^{\text{learn}}$. In this way, the refined learnable class text feature $\boldsymbol{f}^{\text{refined}}$ can be guided by stable class-level semantic priors while maintaining task adaptability

$$\boldsymbol{A}_{\text{dis}} = \sigma\left(\boldsymbol{W}_2^{\text{dis}}\,\text{ReLU}\left(\boldsymbol{W}_1^{\text{dis}}\boldsymbol{f}^{\text{description}}\right)\right) \tag{15}$$

$$\boldsymbol{f}^{\text{refined}} = \widetilde{\boldsymbol{f}}^{\text{learn}} \odot \boldsymbol{A}_{\text{dis}} \tag{16}$$

where $\boldsymbol{W}_1^{\text{dis}}$ and $\boldsymbol{W}_2^{\text{dis}}$ denote linear mapping layers, $\sigma(\cdot)$ is the Sigmoid function, and $\odot$ represents element-wise multiplication.

By constructing a bidirectional interactive fusion structure, the refined learnable class text feature $\boldsymbol{f}^{\text{refined}}$ and the class description text feature $\boldsymbol{f}^{\text{description}}$ perform complementary information interaction in a shared semantic space. In this way, the stability of class description semantics and the flexibility of learnable prompts can be simultaneously preserved, thereby obtaining the discriminative fusion feature $\boldsymbol{T}^{\text{dis}}$

$$\boldsymbol{Q}_r = \boldsymbol{W}_r^q\boldsymbol{f}^{\text{refined}}, \quad \boldsymbol{Q}_d = \boldsymbol{W}_d^q\boldsymbol{f}^{\text{description}} \tag{17}$$

$$\boldsymbol{K}_r = \boldsymbol{W}_r^k\boldsymbol{f}^{\text{refined}}, \quad \boldsymbol{K}_d = \boldsymbol{W}_d^k\boldsymbol{f}^{\text{description}} \tag{18}$$

$$\boldsymbol{V}_r = \boldsymbol{W}_r^v\boldsymbol{f}^{\text{refined}}, \quad \boldsymbol{V}_d = \boldsymbol{W}_d^v\boldsymbol{f}^{\text{description}} \tag{19}$$

$$\boldsymbol{T}^{\text{dis}} = \widetilde{\boldsymbol{f}}^{\text{learn}} + \text{LayerNorm}\left(\boldsymbol{W}_f\left[\sigma\left(\boldsymbol{Q}_d \odot \boldsymbol{K}_r\right) \odot \boldsymbol{V}_r \,\|\, \sigma\left(\boldsymbol{Q}_r \odot \boldsymbol{K}_d\right) \odot \boldsymbol{V}_d\right]\right) \tag{20}$$

where $\boldsymbol{W}_f$ denotes the linear mapping parameter of the fusion layer, $\|$ represents feature concatenation, and $\boldsymbol{W}_r^q$, $\boldsymbol{W}_r^k$, $\boldsymbol{W}_r^v$ and $\boldsymbol{W}_d^q$, $\boldsymbol{W}_d^k$, $\boldsymbol{W}_d^v$ denote the projection matrices that map the features $\boldsymbol{f}^{\text{refined}}$ and $\boldsymbol{f}^{\text{description}}$ into the query, key, and value spaces, respectively.

4) Cross-Modal Similarity-Guided Learning Module: To achieve visual-semantic alignment, the CSLM maps regional visual features into the semantic space, as illustrated in Fig. 5. Since novel-class visual features are easily biased toward base-class patterns under limited supervision, semantic priors can provide category-level guidance independent of visual sample scarcity. By aligning visual features with semantic features, CSLM encourages visual features to focus on target-related semantic cues, thereby reducing base-class bias and enhancing novel-class representation capability. Let $\widetilde{\boldsymbol{R}} \in \mathbb{R}^{N\times M}$ denote the class-aware visual features of N RoIs, and let $\boldsymbol{T}^{\text{dis}} \in \mathbb{R}^{C\times d}$ denote the discriminative semantic features of C classes. The visual features are first projected into the d-dimensional semantic space to obtain $\boldsymbol{V} \in \mathbb{R}^{N\times d}$. The cosine similarity between each RoI and all class-level semantic features is then computed to construct the visual-semantic relation matrix $\boldsymbol{S} \in \mathbb{R}^{N\times C}$

$$\boldsymbol{V} = \widetilde{\boldsymbol{R}}\boldsymbol{W}_V^T \tag{21}$$

$$s_{i,j} = \frac{\boldsymbol{V}_i \cdot \boldsymbol{T}_j^{\text{dis}}}{\|\boldsymbol{V}_i\|\|\boldsymbol{T}_j^{\text{dis}}\|} \tag{22}$$

$$\boldsymbol{R}^{\text{final}} = \widetilde{\boldsymbol{R}} + \text{ReLU}\left(\text{LayerNorm}\left(\left(\widetilde{\boldsymbol{R}}\,\|\,\boldsymbol{S}\right)\boldsymbol{W}_R^T\right)\right) \tag{23}$$

where $\boldsymbol{W}_V^T \in \mathbb{R}^{M\times d}$ denotes the visual projection matrix. Each element $s_{i,j}$ measures the semantic relationship between the ith RoI and the jth class, and the ith row of $\boldsymbol{S}$ therefore serves as an RoI-specific semantic relation representation over all classes. The relation matrix $\boldsymbol{S}$ is concatenated with $\widetilde{\boldsymbol{R}}$ along the feature dimension, producing a feature of size $N \times (M + C)$. The concatenated feature is subsequently projected back to the original M-dimensional visual space through $\boldsymbol{W}_R^T \in \mathbb{R}^{(M+C)\times M}$. Since LayerNorm and ReLU preserve the feature dimensionality, the resulting feature can be combined with $\widetilde{\boldsymbol{R}}$ through a residual connection to obtain $\boldsymbol{R}^{\text{final}} \in \mathbb{R}^{N\times M}$.

Meanwhile, based on the visual–semantic relation matrix, the VSCA strengthens semantic-guided novel-class representation learning by aligning visual features with their corresponding semantic priors. Specifically, VSCA encourages novel-class visual features to move closer to matched semantic representations while keeping them away from unmatched semantic representations. In this way, novel-class visual features become less dependent on base-class visual patterns and obtain stronger category discriminability

$$L_{VSCA} = \frac{1}{N}\sum_{i=1}^{N} -\log\frac{\exp\left(s_{i,y_i}/\tau\right)}{\sum_{k=1}^{C}\exp\left(s_{i,k}/\tau\right)} \tag{24}$$

where N denotes the number of visual features, τ is the temperature coefficient, and y_i is the ground-truth category label of the ith visual features.

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TABLE I
NOVEL/BASE CLASSES FOR DIFFERENT DATASETS

Data Set	Novel Classes	Base Classes
DIOR Split1 [3]	baseballfield, basketballcourt, bridge, chimney, ship	Rest
DIOR Split2 [3]	airplane, airport, expressway-toll-station, harbor, groundtrackfield	Rest
NWPU VHR-10 [68]	airplane, baseball-diamond, tennis-court	Rest
RSOD Split1 [69]	aircraft	Rest
RSOD Split2 [69]	overpass	Rest

IV. EXPERIMENTS

A. Datasets

1) DIOR: The DIOR dataset serves as a large-scale benchmark for remote sensing object detection. This dataset contains 23 463 remote sensing images and 192 472 annotated object instances, spanning 20 common object categories. Each image has a size of 800×800 pixels. In this article, DIOR is used to evaluate the performance of the proposed framework in large-scale remote sensing scenarios with diverse object categories and complex backgrounds.

2) NWPU VHR-10: The NWPU VHR-10 dataset serves as a representative high-resolution remote sensing benchmark for object detection, covering 10 object categories. This dataset contains 800 remote sensing images, including 650 positive images with annotated object instances and 150 negative images without target objects. In this article, NWPU VHR-10 is adopted to evaluate the detection performance of the proposed framework in small-scale scenarios involving remote sensing imagery with high spatial resolution.

3) RSOD: The RSOD dataset serves as an object detection benchmark for remote sensing imagery with high spatial resolution. This dataset contains 976 optical remote sensing images at a resolution of 640×640 pixels, covering 4 representative object categories. In this article, RSOD is further adopted as a supplementary evaluation dataset to assess the generalization capability of the proposed framework across different remote sensing datasets.

B. Experimental Settings

For the experimental setup, we follow the configurations adopted in previous studies [29], [33], [68], [70], as summarized in Table I. For the DIOR dataset, 15 categories are assigned to the base classes, while the other 5 categories are treated as novel classes. For the NWPU VHR-10 dataset, 7 categories are assigned to the base classes, and the remaining 3 categories are treated as novel classes. For the RSOD dataset, which consists of four object categories, we follow the dataset split protocol proposed in InfRS [35], where one category is selected as the novel class, and the other three are used as base classes.

The SCNet method is built based on SAE-FSDet and adopts ResNet101 with an FPN as the backbone network. All experiments are conducted on a single NVIDIA A100-SXM4 GPU. In both the base-training and fine-tuning phases, optimization is performed using stochastic gradient descent (SGD), with the momentum and weight decay set to 0.9 and 1e-4, respectively. For base training, the model is optimized for 18 epochs, where the learning rate is initialized to 0.001 for the first 12 epochs and then decreased to 0.0001 for the remaining epochs. In the fine-tuning phase, training iterations are configured based on task-specific requirements. For evaluation, mean average precision (mAP) is used to measure detection performance, following the Pascal VOC2007 benchmark protocol for mAP computation. During inference, CAM is bypassed because its prototype-based relation modeling relies on ground-truth labels, while DSM remains active for semantic refinement and classification. The confidence score threshold and IoU threshold used in nonmaximum suppression (NMS) are configured as 0.05 and 0.5, respectively. SCNet employs a frozen CLIP ViT-B/32 as the text encoder. The learnable prompts consist of 16 shared 512-D vectors that are randomly initialized and prepended to the class-name embedding of each category to form the complete learnable prompt sequence. During training, the CLIP text encoder and class-name embeddings remain frozen, while only the learnable prompts and the subsequent semantic adaptation and fusion modules are updated. The loss hyperparameters are set to $\alpha = 0.5$ and $\beta = 0.2$, the margin parameter of the AMS loss is set to $m = 0.15$, and the temperature parameter of the VSCA loss is set to $\tau = 0.07$. The loss settings in the base-training stage remain consistent with those of the baseline. In addition, the predefined class-description prompts used in the text modality follow the design strategy of Lu et al. [38] and remain fixed across all shot settings and experimental configurations.

C. Experimental Results and Comparison

1) Results on the DIOR Dataset: Table II presents the comparison results between the proposed SCNet method and several state-of-the-art methods on the DIOR dataset. From the overall results, SCNet achieves competitive and even superior performance under most experimental settings. Specifically, among the eight experimental settings, our method obtains the best results in six settings and also achieves second-best performance in the remaining settings, demonstrating overall superiority over most compared methods.

In Split1, compared with the advanced method proposed by Xu et al. [72], SCNet improves by 3.65%, 1.62%, 1.07%, and 0.60% from the 3-shot to 20-shot settings, respectively. Moreover, compared with SAE-FSDet, it achieves improvements of 4.30%, 3.17%, 1.82%, and 1.97%, respectively. In Split2, the proposed method maintains an overall performance level comparable to leading competing methods, ranking first in the 3-shot and 20-shot settings, while obtaining the second-highest results in the 5-shot and 10-shot settings. According to the averaged results, SCNet ranks first across all four shot settings.

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TABLE II

COMPARISON OF DIFFERENT FSOD METHODS ON DIOR WITH K-SHOT CONFIGURATIONS ($K = 3, 5, 10, 20$). ALL REPORTED VALUES ARE MAP (%) FOR NOVEL CLASSES. RED AND BLUE INDICATE THE HIGHEST AND SECOND-HIGHEST RESULTS, RESPECTIVELY. "AVG." DENOTES THE AVERAGE PERFORMANCE ACROSS SPLITS. * DENOTES THE RESULTS REPORTED AS MEAN ± STANDARD DEVIATION OVER THREE INDEPENDENTLY SAMPLED SUPPORT SETS, WHERE ALL COMPARED METHODS ARE EVALUATED USING IDENTICAL SUPPORT-SET CONFIGURATIONS

Split	Method	Publish	3-shot	5-shot	10-shot	20-shot
1	Meta-RCNN [42]	ICCV2019	12.02	13.09	14.07	14.45
	TFA w/cos [31]	ICML2020	16.07	15.36	16.45	18.93
	FsDetView [15]	ECCV2020	13.19	14.29	18.02	18.01
	FSOD [22]	CVPR2021	15.94	20.27	24.22	28.16
	FSCE [26]	CVPR2021	27.91	28.60	33.05	37.55
	MSOCL [56]	TGRS2022	24.97	27.27	33.37	39.22
	G-FSDet [32]	ISPRS2023	27.57	30.52	37.46	39.83
	ICPE [71]	AAAI2023	11.68	12.34	12.95	14.33
	VFA [44]	AAAI2023	21.94	21.27	23.32	24.28
	SAE-FSDet [33]	TGRS2024	28.80	32.40	37.09	42.46
	Xu et al. [72]	TGRS2025	29.45	33.95	37.84	43.83
	SCNet	-	33.10	35.57	38.91	44.43
	MSOCL* [56]	TGRS2022	21.69 ± 4.48	28.17 ± 1.18	34.66 ± 0.92	38.23 ± 0.76
	G-FSDet* [32]	ISPRS2023	26.50 ± 0.77	29.38 ± 0.82	35.73 ± 1.28	37.93 ± 1.35
	SAE-FSDet* [33]	TGRS2024	29.29 ± 0.87	32.20 ± 1.91	37.69 ± 0.97	44.28 ± 1.06
	SCNet*	-	31.53 ± 1.19	33.89 ± 1.63	38.80 ± 0.29	44.98 ± 0.39
2	Meta-RCNN [42]	ICCV2019	8.84	10.88	14.90	16.71
	TFA w/cos [31]	ICML2020	6.81	7.53	8.93	11.05
	FsDetView [15]	ECCV2020	10.83	9.63	13.57	14.76
	FSOD [22]	CVPR2021	9.35	9.73	14.84	16.20
	FSCE [26]	CVPR2021	13.17	14.07	15.79	20.93
	MSOCL [56]	TGRS2022	13.31	13.40	15.00	18.15
	G-FSDet [32]	ISPRS2023	14.13	15.84	15.20	22.69
	ICPE [71]	AAAI2023	10.92	10.56	12.39	13.18
	VFA [44]	AAAI2023	12.10	12.70	14.72	15.47
	SAE-FSDet [33]	TGRS2024	13.99	15.65	17.41	21.34
	Xu et al. [72]	TGRS2025	14.72	16.72	18.85	23.02
	SCNet	-	15.41	16.40	18.42	23.16
	MSOCL* [56]	TGRS2022	10.11 ± 3.67	14.15 ± 0.57	18.21 ± 2.36	23.50 ± 1.71
	G-FSDet* [32]	ISPRS2023	12.19 ± 1.94	14.66 ± 0.85	17.02 ± 2.17	22.00 ± 1.03
	SAE-FSDet* [33]	TGRS2024	13.36 ± 2.72	14.08 ± 1.66	19.30 ± 1.80	23.93 ± 2.02
	SCNet*	-	15.30 ± 2.52	16.88 ± 2.60	20.53 ± 1.56	25.17 ± 1.53
Avg.	Meta-RCNN [42]	ICCV2019	10.43	11.99	14.49	15.58
	TFA w/cos [31]	ICML2020	11.44	11.45	12.69	14.99
	FsDetView [15]	ECCV2020	12.01	11.96	15.80	16.39
	FSOD [22]	CVPR2021	12.65	15.00	19.53	22.18
	FSCE [26]	CVPR2021	20.54	21.34	24.42	29.24
	MSOCL [56]	TGRS2022	19.14	20.34	24.19	28.69
	G-FSDet [32]	ISPRS2023	20.85	23.18	26.33	31.26
	ICPE [71]	AAAI2023	11.30	11.45	12.67	13.76
	VFA [44]	AAAI2023	17.02	16.99	19.02	19.88
	SAE-FSDet [33]	TGRS2024	21.40	24.03	27.25	31.90
	Xu et al. [72]	TGRS2025	22.09	25.34	28.35	33.43
	SCNet	-	24.26	25.99	28.67	33.80

These results indicate that SCNet achieves favorable performance across different data splits and shows clear advantages in low-shot scenarios. Furthermore, when evaluated over three independently sampled support sets using identical support configurations across methods, SCNet still achieves the best overall performance, further demonstrating its robustness to support-sample variation.

For a more intuitive analysis, beyond quantitatively verifying SCNet through evaluation metrics, we also provide qualitative visualizations. Fig. 6 illustrates the visualization outputs for the DIOR test set with the 3-shot configuration in the first data split. The visual comparisons indicate that, relative to the baseline, SCNet still shows favorable detection capability for novel classes even in the most challenging experimental setting.

2) Results on the NWPU VHR-10 Dataset: Compared with DIOR, NWPU VHR-10 covers fewer target categories, with relatively limited images and annotated object instances for each category. Such characteristics of smaller dataset scale and sparse sample distribution make the supervision information available during training more limited, thereby posing greater challenges to FSOD tasks. Therefore, the experimental results obtained on NWPU VHR-10 can more effectively reflect SCNet's learning capacity in low-shot scenarios, as well as its adaptability to novel classes.

Table III summarizes the quantitative comparisons among different methods on the NWPU VHR-10 dataset. Overall, SCNet demonstrates strong competitiveness against a range of leading methods, achieving either the best or second-best performance across different shot settings. In particular, its

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Fig. 6. Detection visualization for Split1 of the DIOR test set in the 3-shot setting. Novel classes: *baseballfield*, *basketballcourt*, *bridge*, *chimney*, and *ship*.

TABLE III

COMPARISON OF DIFFERENT FSOD METHODS ON NWPU VHR-10 WITH K-SHOT CONFIGURATIONS ($K = 3, 5, 10, 20$). ALL REPORTED VALUES ARE MAP (%) FOR NOVEL CLASSES. RED AND BLUE INDICATE THE HIGHEST AND SECOND-HIGHEST RESULTS, RESPECTIVELY. * DENOTES THE RESULTS REPORTED AS MEAN ± STANDARD DEVIATION OVER THREE INDEPENDENTLY SAMPLED SUPPORT SETS, WHERE ALL COMPARED METHODS ARE EVALUATED USING IDENTICAL SUPPORT-SET CONFIGURATIONS

Method	Publish	3-shot	5-shot	10-shot	20-shot
Meta-RCNN [42]	ICCV2019	20.51	21.77	26.98	28.24
TFA w/cos [31]	ICML2020	16.17	20.49	21.22	21.57
P-CNN [29]	TGRS2021	41.80	49.17	63.29	66.83
FSCE [26]	CVPR2021	41.63	48.80	59.97	79.60
MSOCL [56]	TGRS2022	45.00	49.04	63.27	78.43
G-FSDet [32]	ISPRS2023	49.05	56.10	71.82	75.41
SAE-FSDet [33]	TGRS2024	57.96	59.40	71.02	85.08
RP-FSOD [60]	TCSVT2024	50.45	60.72	72.91	81.46
ST-FSOD [37]	TGRS2025	51.59	58.49	72.75	79.46
RIFR [59]	TGRS2025	53.17	62.39	72.29	77.01
SCG-FSOD [66]	TGRS2026	51.52	64.95	75.99	82.21
SCNet	-	65.41	66.94	74.52	85.19
MSOCL* [56]	TGRS2022	39.15 ± 7.71	49.06 ± 4.07	60.76 ± 2.24	67.71 ± 7.92
G-FSDet* [32]	ISPRS2023	38.61 ± 7.39	49.10 ± 5.37	57.86 ± 9.87	62.71 ± 9.04
SAE-FSDet* [33]	TGRS2024	52.02 ± 8.26	56.00 ± 4.55	71.92 ± 2.22	78.40 ± 4.26
SCNet*	-	55.63 ± 9.56	63.15 ± 3.35	74.02 ± 2.63	79.19 ± 4.24

advantages are more pronounced under low-shot conditions. Under the more challenging 3-shot and 5-shot settings, SCNet achieves mAP values of 65.41% and 66.94%, respectively. Compared with the recent leading methods SCG-FSOD and RIFR, SCNet outperforms SCG-FSOD by 13.89 and 1.99 percentage points, and RIFR by 12.24 and 4.55 percentage points, respectively. Meanwhile, it improves upon SAE-FSDet by 7.45 and 7.54 percentage points, further demonstrating its stronger detection capability when supervision is extremely limited.

Compared with the conventional Meta-RCNN and TFA w/cos methods, SCNet improves the mAP over Meta-RCNN by 44.90, 45.17, 47.54, and 56.95 percentage points under the 3-shot, 5-shot, 10-shot, and 20-shot settings, respectively, and by 49.24, 46.45, 53.30, and 63.62 percentage points over TFA w/cos. Meta-RCNN and TFA w/cos mainly perform independent prediction for individual RoIs, with limited exploitation of interclass relationships and semantic information. In contrast, SCNet collaboratively refines RoI representations through class-relation modeling and dynamic semantic guidance, thereby producing more discriminative object features when training samples are limited and categories are prone to confusion. In addition, the evaluation using independently sampled support sets reveals noticeable performance variations caused by support-sample selection. Nevertheless, when all methods are evaluated under identical support-set configurations, SCNet still achieves the best overall performance, demonstrating its robustness to support-set variation. Qualitative results under the most challenging 3-shot setting are further presented in Fig. 7. The visual comparisons show that SCNet can accurately localize and recognize novel-class objects while maintaining reliable detection performance.

3) Results on the RSOD Dataset: Due to the absence of a unified split protocol for the RSOD dataset, we follow the setting of Xu et al. [72] for the experiments and use RSOD as an additional validation benchmark to assess SCNet's

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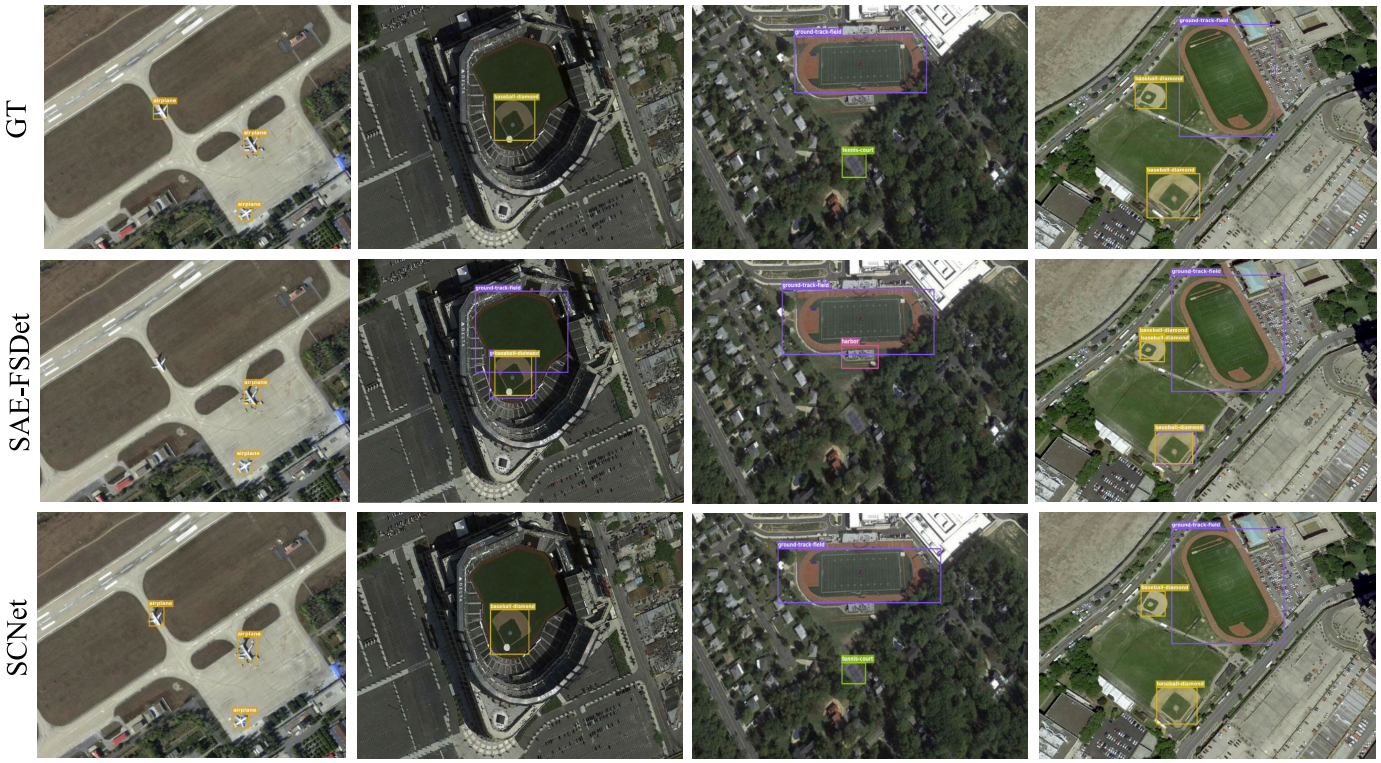


Fig. 7. Detection visualization for the NWPU VHR-10 test set in the 3-shot setting. Novel classes: *airplane*, *baseball-diamond*, and *tennis-court*.

TABLE IV

COMPARISON OF DIFFERENT FSOD METHODS ON RSOD WITH K-SHOT CONFIGURATIONS ($K = 3, 5, 10$). ALL REPORTED VALUES DENOTE MAP (%) FOR NOVEL CLASSES. RED AND BLUE INDICATE THE HIGHEST AND SECOND-HIGHEST RESULTS, RESPECTIVELY. "AVG." DENOTES THE AVERAGE PERFORMANCE ACROSS SPLITS. * DENOTES THE RESULTS REPORTED AS MEAN ± STANDARD DEVIATION OVER THREE INDEPENDENTLY SAMPLED SUPPORT SETS, WHERE ALL COMPARED METHODS ARE EVALUATED USING IDENTICAL SUPPORT-SET CONFIGURATIONS

Split	Method	Publish	3-shot	5-shot	10-shot
1	MSOCL [56]	TGRS2022	14.48	14.64	16.19
	G-FSDet [32]	ISPRS2023	9.09	9.95	15.19
	SAE-FSDet [33]	TGRS2024	10.80	19.30	18.80
	SCNet	-	22.70	21.50	29.50
	MSOCL* [56]	TGRS2022	15.28 ± 0.74	17.38 ± 2.11	19.30 ± 2.37
	G-FSDet* [32]	ISPRS2023	9.67 ± 0.81	9.45 ± 0.82	18.39 ± 3.36
	SAE-FSDet* [33]	TGRS2024	12.17 ± 5.43	17.87 ± 1.01	29.03 ± 8.03
	SCNet*	-	18.80 ± 4.24	20.80 ± 0.57	36.87 ± 6.21
2	MSOCL [56]	TGRS2022	29.53	32.79	41.82
	G-FSDet [32]	ISPRS2023	19.76	18.09	23.56
	SAE-FSDet [33]	TGRS2024	37.60	39.60	41.50
	SCNet	-	38.70	40.20	44.10
	MSOCL* [56]	TGRS2022	24.42 ± 3.63	26.56 ± 4.43	39.88 ± 8.49
	G-FSDet* [32]	ISPRS2023	19.82 ± 0.23	22.62 ± 3.42	36.25 ± 12.31
	SAE-FSDet* [33]	TGRS2024	27.80 ± 6.93	31.07 ± 6.09	41.40 ± 9.19
	SCNet*	-	32.80 ± 5.17	33.80 ± 4.73	46.23 ± 9.35
Avg.	MSOCL [56]	TGRS2022	22.01	23.72	29.01
	G-FSDet [32]	ISPRS2023	14.43	14.02	19.38
	SAE-FSDet [33]	TGRS2024	24.20	29.45	30.15
	SCNet	-	30.70	30.85	36.80

generalizability. Table IV summarizes the results, where SCNet delivers competitive performance across different splits. Based on the averaged results, SCNet improves upon SAE-FSDet by approximately 6.50, 1.40, and 6.65 percentage points under the 3-shot, 5-shot, and 10-shot settings, respectively. The additional evaluations with different support sets further show that RSOD exhibits relatively large performance fluctuations, especially under some low-data configurations. Even under such variations, SCNet maintains clear advantages over the compared methods, with particularly notable gains in the

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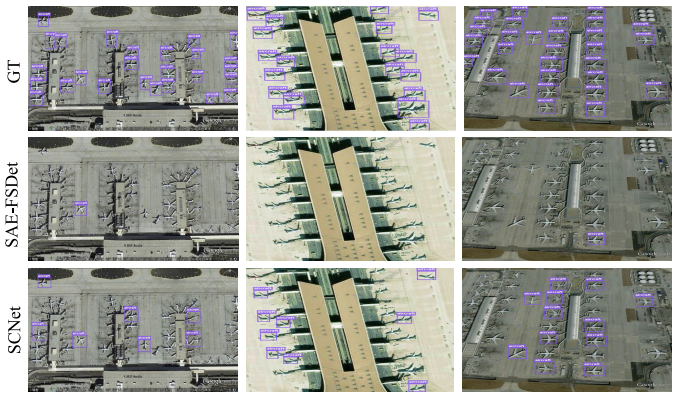


Fig. 8. Detection visualization for the RSOD test set in the 3-shot setting of Split1. Novel-class: *aircraft*.

TABLE V
ABLATION STUDY OF EACH COMPONENT IN SCNET. ALL REPORTED RESULTS REPRESENT THE MAP (%) ON NOVEL CLASSES. BOLD INDICATES THE HIGHEST RESULTS

	Baseline	CAM	DSM	3-shot	5-shot
I	✓			57.96	59.40
II	✓	✓		60.15	64.88
III	✓		✓	59.29	63.02
IV	✓	✓	✓	**65.41**	**66.94**

3-shot and 10-shot settings. Meanwhile, Fig. 8 presents the detection visualizations for Split1 under the 3-shot setting.

D. Ablation Study

To assess the contribution of each module in SCNet, extensive ablation studies are performed. All experiments reported in this part are carried out using the NWPU VHR-10 dataset.

1) Ablation of SCNet Components: SCNet consists of two core components, CAM and DSM. To evaluate the effectiveness of each component and their synergistic interaction, we conduct component-wise ablation experiments, as reported in Table V. The baseline model achieves mAP values of 57.96% and 59.40% under the 3-shot and 5-shot settings, respectively. When CAM is introduced individually, the performance increases to 60.15% and 64.88%, corresponding to gains of 2.19% and 5.48%. This demonstrates that modeling the contextual relationships among class prototypes and encoding the relative relationships between the target class and its confusing classes into RoI features can effectively enhance the class discriminability of visual representations. When DSM is applied alone, the mAP reaches 59.29% and 63.02%, yielding improvements of 1.33% and 3.62%, respectively. These results indicate that the dynamic semantic priors constructed from learnable prompts and class-description prompts can supplement the limited visual information available under few-shot conditions and alleviate base-class bias in novel-class representations to some extent.

When CAM and DSM are jointly employed, the model achieves the best mAP values of 65.41% and 66.94% under the 3-shot and 5-shot settings, outperforming the baseline by 7.45% and 7.54%, respectively. These improvements arise from the sequential collaboration between the two modules rather than from their independent effects. CAM first strengthens the response of each RoI to the ground-truth class prototype while suppressing interference from confusing classes, thereby providing DSM with relation-aware visual features characterized by clearer class boundaries. DSM then projects these features into the semantic space and performs matching, alignment, and fusion with dynamic semantic priors. In this way, the class-relation information extracted by CAM directly affects the matching distribution between each RoI and different class semantics, reducing the probability of high-response matching with incorrect categories. On this basis, DSM further supplements the visual representations with stable yet task-adaptive semantic knowledge and corrects residual category confusion. These results demonstrate that CAM provides a more discriminative visual foundation for semantic matching in DSM, while DSM further enhances the classification capability of the CAM-refined features, confirming the complementarity and synergy between the two modules.

TABLE VI
ABLATION STUDY OF CAM IN THE SCNET FRAMEWORK. ALL REPORTED RESULTS REPRESENT THE MAP (%) ON NOVEL CLASSES. BOLD INDICATES THE HIGHEST RESULTS

Method	3-shot	5-shot	10-shot	20-shot
SCNet w. RPA	61.09	62.94	71.20	83.86
SCNet w. ARPA	63.42	65.23	73.04	85.07
SCNet w. REPA	61.13	63.97	71.62	84.52
SCNet	**65.41**	**66.94**	**74.52**	**85.19**

2) Ablation Study of CAM: To further analyze the roles of the individual components within CAM and their synergistic relationships, we conduct a progressive ablation study, with the results presented in Table VI. Four variants are considered: RPA, ARPA, REPA, and the complete CAM. RPA performs attention interaction only between ordinary class prototypes and RoI features. ARPA introduces the AMS loss on top of RPA to constrain the attention distribution. REPA employs class prototypes enhanced by prototype self-attention to refine RoI features, but does not use the AMS constraint. The complete CAM combines both relation-enhanced prototypes and the AMS loss. Under the 3-shot, 5-shot, 10-shot, and 20-shot settings, RPA achieves mAP values of 61.09%, 62.94%, 71.20%, and 83.86%, respectively, indicating that explicit RoI–prototype interaction can effectively introduce class information into RoI representations. After incorporating AMS, ARPA improves the performance to 63.42%, 65.23%, 73.04%, and 85.07%, demonstrating that the constraint strengthens the response of RoI features to the ground-truth class prototype while suppressing interference from visually similar hard-negative classes. REPA obtains mAP values of 61.13%, 63.97%, 71.62%, and 84.52%, respectively, validating the effectiveness of class-prototype relation modeling in enhancing the class awareness of RoI features. The complete CAM achieves the best results across all shot settings, reaching 65.41%, 66.94%, 74.52%, and 85.19%, respectively. These results demonstrate the clear

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TABLE VII

ABLATION STUDY OF THE NUMBER OF HEADS H IN CAM. ALL REPORTED RESULTS REPRESENT THE MAP (%) ON NOVEL CLASSES. BOLD INDICATES THE HIGHEST RESULTS

Number of Heads H	1	**2**	4	8
3-shot	61.08	**65.41**	64.37	63.50

TABLE VIII

ABLATION STUDY OF THE FUSION RATIO FACTOR λ IN CAM. ALL REPORTED RESULTS REPRESENT THE MAP (%) ON NOVEL CLASSES. BOLD INDICATES THE HIGHEST RESULTS

Fusion Ratio Factor λ	0.1	**0.2**	0.3	0.4
3-shot	62.65	**65.41**	64.42	62.44

TABLE IX

ABLATION STUDY OF DSM. ALL REPORTED RESULTS REPRESENT THE MAP (%) ON NOVEL CLASSES. BOLD INDICATES THE HIGHEST RESULTS

Method	**3-shot**	**5-shot**	**10-shot**	**20-shot**
SCNet w/o LCP	58.20	64.77	73.98	85.05
SCNet w/o CDP	62.90	65.91	73.69	84.72
SCNet w/o DBA	57.24	64.42	74.18	85.05
SCNet w/o DA	60.46	64.74	73.95	85.03
SCNet w/o VSCA	63.07	65.47	72.86	84.32
SCNet w. GP	64.03	66.39	73.23	84.31
SCNet w. RSP	64.71	65.27	73.53	84.68
SCNet	**65.41**	**66.94**	**74.52**	**85.19**

complementarity between relation-enhanced prototypes and the AMS constraint. Their joint use enables more effective exploitation of class-relation information and produces more discriminative RoI representations, especially under extremely limited training samples.

3) Ablation Study on the Parameters in CAM: In CAM, the multihead attention mechanism is employed to capture relational cues across distinct feature subspaces. Table VII reports the ablation results with respect to the attention head number H. When $H = 2$, the model obtains optimal performance, indicating that a suitable multihead configuration can enhance relational interaction modeling. However, further enlarging the head number yields no extra gains, which may be because excessive partitioning of feature subspaces weakens effective information modeling under few-shot supervision. In addition, the ablation results for the fusion weight factor λ are shown in Table VIII. These results indicate that when $\lambda = 0.2$, the model reaches optimal performance, suggesting that this setting can provide a better tradeoff between original feature stability and enhanced feature utilization. Therefore, H is finally set to 2, and λ is set to 0.2.

4) Ablation Study of DSM: To assess the role of each DSM component, ablation studies are performed, and the results are reported in Table IX. Among them, LCP, CDP, DBA, DA, and VSCA denote learnable class prompts, class-description prompts, the DBA, discriminative activation, and the VSCA, respectively. Overall, removing any component leads to performance degradation, indicating that each part of DSM plays a positive role in novel-class representation learning.

From the perspective of semantic prior construction, LCP and CDP provide task-adaptive semantics and stable category semantic constraints, respectively, jointly forming more complete dual-path semantic priors. DBA helps alleviate the gap between generic textual semantics and remote sensing object semantics, making the generated semantic features more suitable for FSOD in remote sensing imagery. DA further strengthens the discriminative interaction between the two types of semantic prompts, enabling task-adaptive semantics to be guided by stable category semantic priors and thus generating more discriminative dynamic semantic priors. Furthermore, VSCA aligns visual features with semantic priors, enabling semantic information to effectively guide visual feature learning and reduce the dependence of novel-class representations on base-class visual patterns. When all components are used together, SCNet delivers the strongest performance across all shot settings, achieving 65.41%, 66.94%, 74.52%, and 85.19%, respectively. This further demonstrates that the dual-path co-activated semantics constructed by DSM can effectively guide visual feature learning.

Based on the ablation results of GP and RSP, we further analyze the influence of different textual prompt formulations on model performance. GP and RSP denote the generic template prompt for natural scenes and the generic template prompt for remote sensing scenes, respectively. The results show that the complete SCNet consistently outperforms the variants using GP or RSP across all shot settings. This indicates that, compared with generic templates or prompts that simply introduce remote sensing scene information, class-description prompts containing category-specific attribute, structural, and functional semantics provide more discriminative semantic constraints. Notably, under the 5-shot setting, RSP achieves an mAP of 65.27%, which is lower than the 66.39% obtained by GP. This result suggests that merely adding scene-level descriptions does not necessarily yield stable performance gains, whereas fine-grained semantic information directly related to category discrimination is more important.

TABLE X

ABLATION STUDY ON THE NUMBER OF LEARNABLE PROMPT VECTORS IN DSM. ALL REPORTED RESULTS REPRESENT THE MAP (%) ON NOVEL CLASSES. BOLD INDICATES THE HIGHEST RESULTS

Number of Learnable Prompt Vectors	8	12	**16**	20
3-shot	60.37	62.59	**65.41**	59.88

5) Ablation Study of Learnable Prompt Vectors in DSM: We investigate how the quantity of learnable prompt vectors in prompt learning affects model performance and perform ablation experiments in the 3-shot setting, with the results reported in Table X. As the prompt vector quantity increases from 8 to 12 and 16, the model performance gradually rises to 60.37%, 62.59%, and 65.41%, respectively, indicating that appropriately increasing the number of prompt vectors helps enhance

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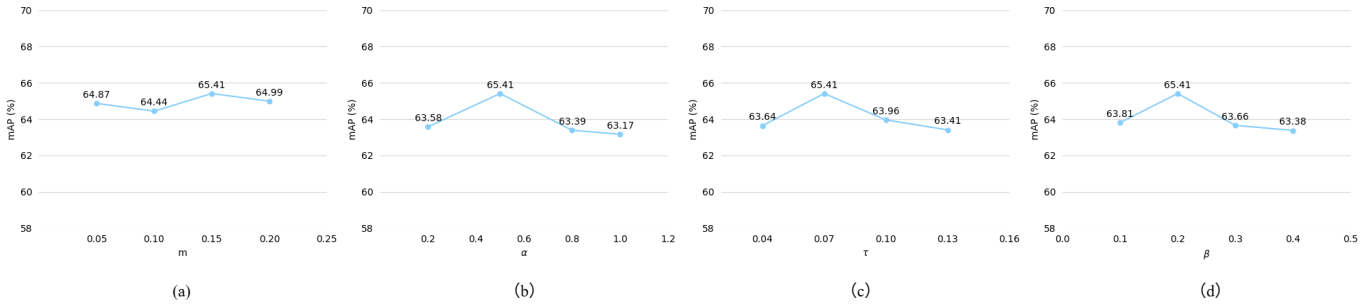


Fig. 9. Ablation of each parameter in the AMS and VSCA loss functions under the 3-shot setting. (a) Margin parameter m in AMS. (b) Loss weight α of the AMS. (c) Temperature parameter τ in VSCA. (d) Loss weight β in VSCA.

the expressive capacity of text prompts, thereby obtaining richer class semantic representations. However, when the number further increases to 20, the performance drops to 59.88%, suggesting that in few-shot scenarios, excessive prompt vectors introduce redundant information and increase optimization difficulty, which is not conducive to model learning. Therefore, the number of prompt vectors is finally set to 16.

6) Ablation Study of Loss Function Parameters: We conduct ablation analysis on the margin parameter m and its weight in the AMS, as shown in Fig. 9(a) and (b). When $m = 0.15$, the model achieves the best performance, and further increasing it leads to a performance drop, indicating that overly strong margin constraints limit the flexibility of feature representation and weaken effective relational modeling. In terms of the weight, the performance increases from 63.58% to 65.41% as the weight increases from 0.2 to 0.5, demonstrating that moderately strengthening prototype alignment constraints helps improve discriminative ability; however, when the weight further increases to 0.8 and 1, the performance decreases to 63.39% and 63.17%, respectively, reflecting that overly strong constraints introduce negative effects. Therefore, m and its weight are finally set to 0.15 and 0.5, respectively.

In addition, Fig. 9(c) and (d) illustrates the effects of the temperature parameter τ and its weight β in the VSCA. When $\tau = 0.07$, the model achieves the best performance of 65.41%, while both overly large and overly small values lead to performance degradation. For the weight setting, the best performance is obtained when $\beta = 0.2$, and further increasing it reduces performance to 63.38%. Therefore, τ and β are finally set to 0.07 and 0.2, respectively.

E. Comprehensive Analysis

1) Visualization Analysis: To intuitively evaluate the ability of SCNet to focus on target regions and the quality of its feature representations, we present class activation maps, t-SNE feature distributions, and detection results in complex scenes. As shown in Fig. 10, compared with the baseline method SAE-FSDet, the activation responses of SCNet are more concentrated on the actual target regions, while the responses to complex backgrounds and irrelevant areas are substantially reduced. This observation indicates that SCNet can capture more discriminative target-related cues, thereby improving object localization and category recognition under

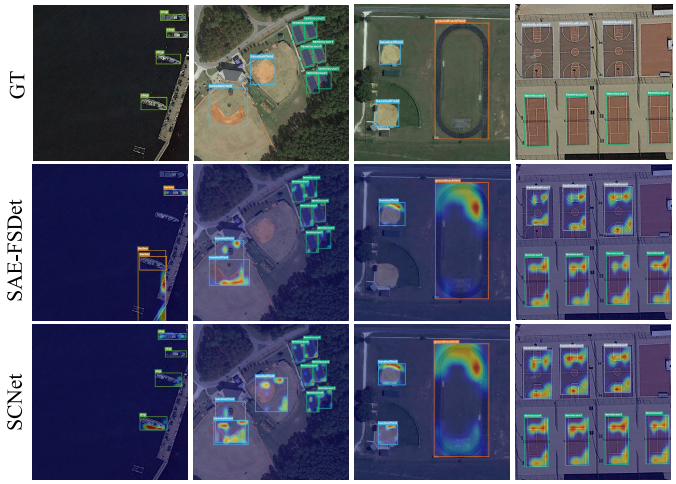


Fig. 10. Activation maps of different object categories under the 3-shot setting on the DIOR, including both novel and base categories.

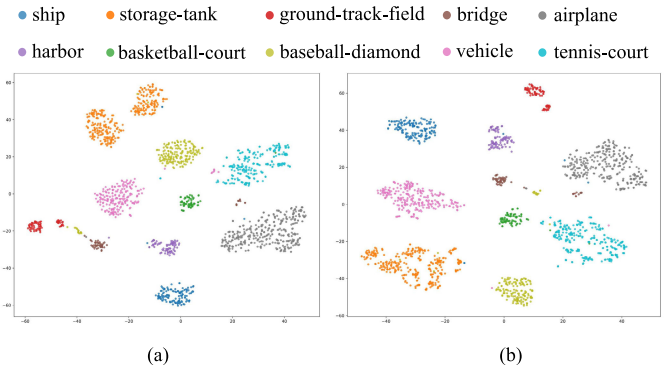


Fig. 11. t-SNE visualization of 300 randomly selected samples from NWPU VHR-10. (a) Baseline and (b) SCNet.

few-shot conditions. For example, SAE-FSDet incorrectly identifies the novel-class object *ship* as *harbor*, and its predicted bounding box exhibits an obvious deviation from the ground-truth target, whereas SCNet achieves more accurate localization and classification.

The t-SNE visualization results in Fig. 11 show that the class features learned by SCNet exhibit clearer interclass boundaries and more compact feature clusters for novel classes, demonstrating that SCNet enhances feature discriminability and reduces confusion between categories with similar visual

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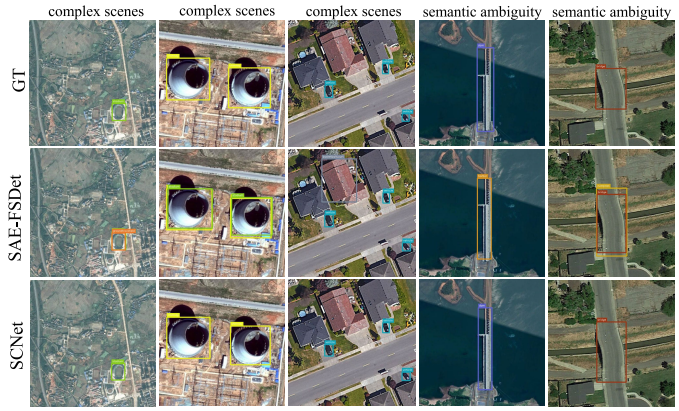


Fig. 12. Visual comparison of detection results for complex and semantically ambiguous categories on the DIOR dataset.

TABLE XI

GENERALIZED FSOD RESULTS ON THE NWPU VHR-10 DATASET. ALL REPORTED VALUES ARE mAP_{50} (%). BOLD INDICATES THE BEST RESULT. * INDICATES THAT THE COMPARED METHODS ARE EVALUATED USING THE SAME SAMPLED SUPPORT SETS UNDER EACH SHOT SETTING

Evaluation	Method	3-shot	5-shot	10-shot	20-shot
Base-class	SAE-FSDet*	**82.85**	82.94	83.32	82.72
	SCNet*	81.74	**83.44**	**84.23**	**84.01**
Novel-class	SAE-FSDet*	58.31	59.93	70.40	84.43
	SCNet*	**65.41**	**66.94**	**74.52**	**85.19**
Overall	SAE-FSDet*	75.49	76.03	79.44	83.23
	SCNet*	**76.84**	**78.49**	**81.32**	**84.36**

appearances. Fig. 12 further compares the detection results of SAE-FSDet and SCNet in complex backgrounds and semantically ambiguous scenes. For sports-facility categories, stadiums, ground track fields, and basketball courts exhibit similar contours and internal structures from the remote-sensing perspective, while some building roofs may also be misidentified as basketball courts because of their similar shapes and colors. Dams and harbors are both commonly located near water bodies and share similar elongated structures and scene contexts. Likewise, bridges and overpasses are prone to category confusion because of their similar elongated crossing structures. SAE-FSDet tends to produce false detections or incorrect classifications in these scenes, whereas SCNet more effectively integrates object structures, class relationships, and semantic information to suppress interference from similar backgrounds and confusing categories. Moreover, in storage-tank scenes with evident shadows and complex backgrounds, SCNet still produces reliable detection results even when parts of the targets are occluded. Overall, CAM enhances the model's ability to distinguish confusing categories by modeling the relationships among class prototypes and incorporating relational information into RoI features. Meanwhile, the dual-prompt semantic priors provided by DSM complement the relation-aware visual features, jointly strengthening discriminative visual cues associated with novel classes.

2) Generalized FSOD Analysis: To further evaluate the balance between novel-class adaptation and base-class retention, we conduct generalized FSOD experiments on the NWPU VHR-10 dataset. As shown in Table XI, SCNet maintains comparable base-class performance to SAE-FSDet, with only a slight decrease under the 3-shot setting and improved performance under the remaining settings. Meanwhile, SCNet consistently improves both novel-class and overall detection performance across different shot settings. These results indicate that the enhancement of novel-class representations is not achieved at the cost of substantial degradation in base-class detection, demonstrating a favorable balance between novel-class adaptation and base-class retention. This balance mainly benefits from the dynamic semantic priors introduced by DSM, which complement limited novel-class visual information while preserving the original visual representations required for base-class recognition.

TABLE XII

TRAINING TIME AND INFERENCE SPEED OF MODELS EVALUATED UNDER THE NWPU VHR-10 5-SHOT SETTING IN THE SAME COMPUTATIONAL ENVIRONMENT. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	MSOCL [56]	SAE-FSDet [33]	SCNet
Inference Speed (FPS)	23.24	**28.06**	23.48
Training Time (h)	**0.67**	0.68	0.76

3) Model Complexity and Efficiency Analysis: To evaluate the model complexity and runtime efficiency of SCNet, we first conducted a unified comparison of the training time and inference speed of MSOCL, SAE-FSDet, and SCNet under the 5-shot setting on the NWPU VHR-10 dataset using the same single NVIDIA A100-SXM4 GPU, as shown in Table XII. SCNet requires 0.76 h for training, which is close to the 0.67 h required by MSOCL and the 0.68 h required by SAE-FSDet. Its inference speed reaches 23.48 FPS, which is slightly higher than the 23.24 FPS of MSOCL but lower than the 28.06 FPS of SAE-FSDet. On this basis, we further compared the model complexity of SCNet with that of several existing methods, as presented in Table XIII. Compared with the baseline model, SCNet incurs increases in both parameter size and computational cost. These additional overheads mainly arise from the introduction of DSM and CAM, which enhance category-relation modeling and visual-semantic interaction while introducing extra parameters and computation. Specifically, SCNet contains 79.85 M parameters, slightly more than FsDetView and G-FSDet, but still within a reasonable range. Its computational cost is 202.25 GFLOPs, which is substantially lower than those of FsDetView and ST-FSOD and remains comparable to those of MSOCL and SAE-FSDet.

Overall, these results indicate that although SCNet introduces additional category-relation modeling and visual-semantic interaction modules, its overall training and inference overhead remains acceptable. Considering the performance improvements brought by these modules, SCNet achieves a favorable balance between detection accuracy and computational efficiency.

4) Limitation Analysis: On the DIOR dataset, SCNet yields improved detection results in most settings. However, since it focuses more on RoI-level local discriminative modeling, its gains may be limited when novel-class decision boundaries

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TABLE XIII
COMPLEXITY AND EFFICIENCY COMPARISON OF DIFFERENT MODELS. BOLD INDICATES THE BEST RESULTS

Method	FsDetView [15]	G-FSDet [32]	MSOCL [56]	SAE-FSDet [33]	ST-FSOD [37]	SCNet
Params (M)	73.96	74.08	61.6	60.89	**58.96**	79.85
GFLOPs	264.56	/	**166.08**	181.13	464.30	202.25

rely more heavily on global scene structures, leading to a few suboptimal cases.

For NWPU VHR-10 with 20 shots, SCNet achieves only limited improvement over the baseline. Existing studies [32], [59] have shown that gains on novel classes in FSOD may be accompanied by some degradation in base-class performance. Since the proposed components are mainly introduced during fine-tuning to enhance novel-class representations, the improvement margin becomes limited when the number of shots increases and novel-class performance approaches the base-class level. Future work will therefore explore strategies that further improve novel-class detection while maintaining stable base-class performance.

Beyond these limitations, improving the generalization of SCNet across different sensors, geographic regions, imaging conditions, and emerging categories remains an important direction. Parameter-efficient or training-free adaptation based on transferable class prototypes, semantic descriptions, and vision–language foundation models may reduce task-specific fine-tuning costs, while lightweight attention and knowledge distillation can further reduce the computational overhead of class-relation interaction and cross-modal alignment for deployment on resource-constrained edge devices.

V. CONCLUSION

We propose SCNet, a semantic cooperative framework for few-shot remote sensing object detection, aiming to enhance novel-class representation learning under limited supervision. SCNet enhances target-related cues in RoI features through contextual relation modeling, and further introduces a dual-prompt-driven semantic guidance mechanism to provide adaptive semantic priors for novel-class representation learning. Through consistency constraints between visual features and semantic information, SCNet forms a synergistic effect between visual context enhancement and semantic knowledge guidance, thereby improving the discriminability and generalization performance of novel-class objects. Extensive experimental results on the DIOR, NWPU VHR-10, and RSOD datasets demonstrate that the proposed method achieves competitive or superior performance compared with state-of-the-art methods under different shot settings. Future work will further explore more efficient semantic guidance learning strategies to reduce the additional computational overhead introduced by the semantic guidance module, while further balancing novel-class detection performance and base-class performance.

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